

LAB-QN-2026-003

NEURAL ORGANOID QUANTUM COHERENCE
EXPERIMENTS

PROJECT: PROJECT LIGHTCONE - Observer Effect Studies

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DATE: January 10-12, 2026
CLASSIFICATION: TOP SECRET // ORACLE EYES ONLY

Experiment Overview

Testing quantum coherence duration in neural organoid substrates under varying temperature and isolation conditions. Hypothesis: Ordered water layers extend coherence time by 1000x compared to free solution.

Day 1: Initial Setup

2026-01-10, 09:30

Prepared 12 organoid samples (batch TM-ORG-2026-A). Each ~4mm diameter, cultured for 45 days. Verified viability via calcium imaging - all samples show spontaneous activity.

Divided into 4 groups (3 samples each):

- Group A: Room temp (22°C), standard medium
- Group B: Cold (4°C), standard medium
- Group C: Room temp, deuterated water
- Group D: Cold (4°C), deuterated water + magnetic shield

Day 1: Baseline Measurements

2026-01-10, 14:15

Coherence measured via ultrafast spectroscopy (100 fs pulse laser).
Baseline results:

Group	Sample 1	Sample 2	Sample 3	Mean ± SD
A (Control)	1.2 ps	1.4 ps	1.1 ps	1.23 ± 0.15 ps
B (Cold)	8.7 ps	9.2 ps	8.1 ps	8.67 ± 0.56 ps
C (D2O)	15.3 ps	16.1 ps	14.8 ps	15.4 ± 0.65 ps
D (Full)	142 ps	156 ps	138 ps	145 ± 9.2 ps

NOTE: Wow! Group D showing 100x improvement over control. Deuterated water + magnetic shielding + cold temp = massive coherence extension. This is way better than predicted.

Day 2: Long-Duration Test

2026-01-11, 10:00

Running extended coherence test on Group D samples. Hypothesis: Can we reach microsecond timescales?

Modified setup: Placed samples in liquid nitrogen-cooled cryostat (77K), maintained magnetic shielding, increased laser pulse repetition for better signal.

11:23 -
Sample D-1 showing coherence oscillations at 1.2 nanoseconds. Still going!

11:45 -

Sample D-1 coherence maintained to 3.7 nanoseconds before decoherence. This is insane. 3000x improvement over control.

12:10 -

Samples D-2 and D-3 similar: 4.1 ns and 3.9 ns respectively. Reproducible! Mean = 3.9 ± 0.2 nanoseconds.

At 3.9 nanoseconds coherence time:

- Neural oscillations: ~ 40 Hz = 25 ms period
- Coherence duration: 3.9 ns
- Ratio: $3.9 \text{ ns} / 25 \text{ ms} = 1.56 \times 10^{-7}$

Still way too short for Orch OR timescale (10-100 ms).

Need another 6-7 orders of magnitude improvement.

BUT: This proves concept. Structured environment CAN extend coherence dramatically. Path forward exists.

Day 3: Temperature Optimization

2026-01-12, 09:00

Testing temperature curve between 77K and 310K to find optimal balance between coherence (favors cold) and biological function (requires warm).

Results (Group D samples, 2-hour incubation at each temp):

Temperature	Coherence Time	Neural Activity
77 K	3.9 ns	None (frozen)
200 K	2.1 ns	None
273 K (0°C)	890 ps	Minimal
280 K (7°C)	420 ps	Weak
290 K (17°C)	180 ps	Moderate
298 K (25°C)	95 ps	Normal
310 K (37°C)	52 ps	Normal

NOTE: Trade-off clear: Colder = better coherence, warmer = better biology. Sweet spot might be 280-285K (7-12°C). Coherence ~400 ps with weak but measurable neural activity.

Conclusions

1. **Structured environment dramatically extends coherence.** Deuterated water + magnetic shielding + cold temp achieved 3000x improvement over baseline. Effect is real and reproducible.
2. **Still insufficient for consciousness timescales.** Even optimized conditions yield nanosecond coherence. Neural processing requires milliseconds. Gap = 6 orders of magnitude.

3. Temperature-biology trade-off is fundamental. Cannot freeze organoids and maintain neural function. Optimal window appears to be 280-290K.

4. Next steps:

- Test additional protective mechanisms (protein scaffolds, quantum error correction?)
- Investigate whether short coherence bursts can accumulate effects
- Explore whether consciousness requires *continuous* coherence or just frequent pulses
- Scale up to larger organoid networks (does size matter?)

NOTE: Personal thought: Even if we can't reach millisecond coherence at biological temps, this proves quantum effects CAN exist in neural tissue. Maybe Orch OR timescale is wrong? Or maybe consciousness doesn't require long coherence, just many rapid quantum events that integrate classically? Need to talk to Dr. Morrison about this. Could have huge implications for the teleportation program.

I certify that this is an accurate record of my work.

Dr. Yuki Tanaka

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